

ARYAN KUMAR

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EDUCATION

RV College of Engineering, Bengaluru

2024 – 2028

B.E. Electronics & Telecommunication Engineering | GPA: 8.85 / 10

Minor in Computer Science – AI

Coursework: Data Structures, Probability & Statistics, Linear Algebra, Calculus

The New Town School, Kolkata

2019 – 2023

Class XII: 85.4% | Class X: 95.2%

EXPERIENCE

Facilities Electrical Intern

Jun 2026 – Present

Chevron Engine, Bengaluru — I&E and Reliability Team

- Built a **load and phase-balancing automation tool** for the Australian Business Unit (ABU): ingests per-feeder load data, computes R/Y/B phase currents and imbalance %, and recommends load redistribution across phases, replacing manual spreadsheet balancing.
- Developed a **control valve sizing calculator** that computes required Cv from flow rate, pressure drop and fluid properties, checks valve opening/rangeability, and standardizes sizing across process loops.
- Support instrumentation work on **process control loops** (transmitter, controller, final control element); perform load calculations and draft & review single-line diagrams (SLDs).
- Assist the reliability team on equipment improvement and optimization initiatives.

Frontend & UI/UX Intern — Procam Group, Kolkata

Jun 2024 – Aug 2024

- Built a responsive UI for a driver app used by **100+ users**; reduced interaction steps by **25%** through optimized workflows.

PROJECTS

Crowd-Sense AI: Multi-Feed Spatial Telemetry Matrix

2026

Tech: Python • YOLOv8 • OpenCV • Streamlit • NumPy • Pandas

- Engineered an **async computer vision pipeline** that ingests multiple camera feeds concurrently, so slow feeds never block real-time object tracking on the others.
- Wrapped **YOLOv8** inference to detect people per frame, convert detections into per-feed crowd counts, and evaluate density saturation against configurable zone capacity.
- Built a **threshold-based alert system** with a live Streamlit dashboard showing per-feed density, alert state, and streaming event logs for operator monitoring.

ML-Driven Smart Energy Monitoring System

2025

Tech: Python • Pandas • Scikit-learn • Streamlit

- Processed **1000+ real-time data points** to enable appliance-level energy tracking; cleaned and aggregated readings into time-series features with Pandas.
- Trained **regression models** to forecast usage, improving estimation accuracy by **15-20%**; built Streamlit dashboards for usage trends and cost insights.

LoRa-Based Physical Layer Security System with PLKG-Inspired Key Generation

2026

Tech: ESP32 • LoRa (SX1278) • OLED • C++ • IoT Security

- Built a three-node wireless system (Alice, Bob, Eve) demonstrating physical-layer security and BB84-inspired shared key generation for IoT networks.
- Developed RSSI-assisted key generation with basis comparison and an OLED interface showing live RSSI, shared key, QBER and link status; reciprocal two-way RSSI achieved **~90% key agreement**.
- Implemented eavesdropper detection via dynamic QBER estimation and threat-level classification.

CSITSS 2026 — Conference Technical Lead, IEEE RVCE

2026

Tech: HTML • CSS • JavaScript • Node.js • TypeScript • React • Next.js • Tailwind CSS

- Developed the official conference website (timeline, registration, paper submission, speakers, venue) with responsive, accessible, cross-device layouts.

PUBLICATIONS

Assessment of Public-Health Implications of 5G Networks and Projected Health-Exposure Considerations for 6G Systems

Aryan Kumar, Niraj Lenka, Krti Pandey, Vartika Mrigwani — International Journal of Advanced Research (IJAR), 2026

TECHNICAL SKILLS

Programming: Python, C, C++, SQL, JavaScript, TypeScript | **Data Science:** Pandas, NumPy, Scikit-learn, TensorFlow, YOLOv8, OpenCV
Visualization: Matplotlib, Seaborn | **Tools:** KiCad, SolidWorks, Jupyter, Git/GitHub, Streamlit

LEADERSHIP & ACTIVITIES

- Student Placement Coordinator (SPC)**, Dept. of E&TC — Class of 2028
- Vice President**, Studio Zero – RVCE filmmaking club
- Featured actor in two short films (2nd Runner-Up, 50+ teams); Top 30 Finalist, SDG Photography Contest; Shooting Guard, RVCE Basketball (inter-college semi-finals & finals)

CERTIFICATIONS

- Data Science for Engineers** — IIT Madras | **Distributed Systems** — IIT Patna